Space exploration

At the start of the 20th century, rockets were invented that were powerful enough to blast away from Earth. By the century's end, thousands of spacecraft and hundreds of people had entered space. The spacecraft of the 21st century are beginning to explore the furthest reaches of our Solar System.

MISSIONS TO SPACE

Space missions have landed people on the Moon and rovers on Mars. They have sampled the atmosphere of Jupiter and explored Saturn, Mercury, and even the Asteroid Belt. These missions help us understand the Solar System and our own planet.

MISSIONS TO THE MOON

NUMBER OF MISSIONS SENT: 100+

KEY MISSIONS:

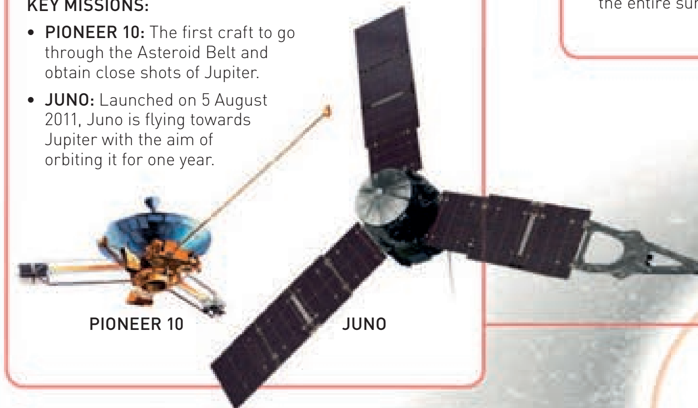
- **APOLLO 15:** Launched in 1971, this was the first of the longer missions, where astronauts stayed for three days on the Moon.
- **LUNAR RECONNAISSANCE ORBITER:** Launched in 2009, this spacecraft is gradually mapping the entire surface of the Moon.

MISSIONS TO JUPITER

NUMBER OF MISSIONS SENT: 9

KEY MISSIONS:

- **PIONEER 10:** The first craft to go through the Asteroid Belt and obtain close shots of Jupiter.
- **JUNO:** Launched on 5 August 2011, Juno is flying towards Jupiter with the aim of orbiting it for one year.



PIONEER 10 JUNO

URANUS

MISSIONS TO ASTEROID BELT

NUMBER OF MISSIONS SENT: 10

KEY MISSIONS:

- **DAWN:** This spacecraft was launched in 2007 to study two bodies in the Asteroid Belt: Vesta and Ceres. It spent a year orbiting Vesta before moving on to Ceres.
- **ROSETTA:** A mission to a comet that photographed asteroids Steins and Lutetia on the way through.

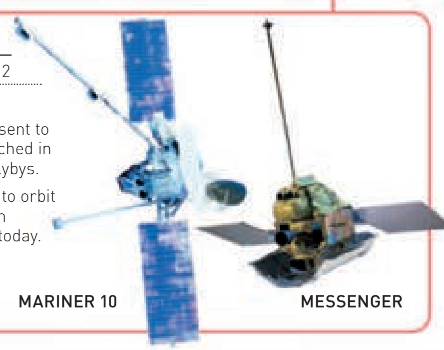
ROSETTA

MISSIONS TO MERCURY

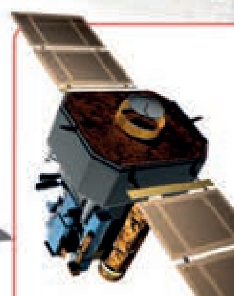
NUMBER OF MISSIONS SENT: 2

KEY MISSIONS:

- **MARINER 10:** The first craft sent to study Mercury, this was launched in 1973 and did three Mercury flybys.
- **MESSENGER:** The first craft to orbit Mercury, this was launched in 2004 and is still in operation today.



MARINER 10 MESSENGER



SOHO

APOLLO MISSION BADGES

The US space programme is run by NASA (National Aeronautics and Space Administration), and it creates a mission patch, or badge, for every space mission. The badges include elements that represent different parts of the mission: its purpose, the name of the space vehicle, and its official number.

APOLLO 1, APOLLO 7, APOLLO 8, APOLLO 9, APOLLO 10, APOLLO 11, APOLLO 12, APOLLO 13, APOLLO 14, APOLLO 15, APOLLO 16, APOLLO 17

THE SPACE AGE

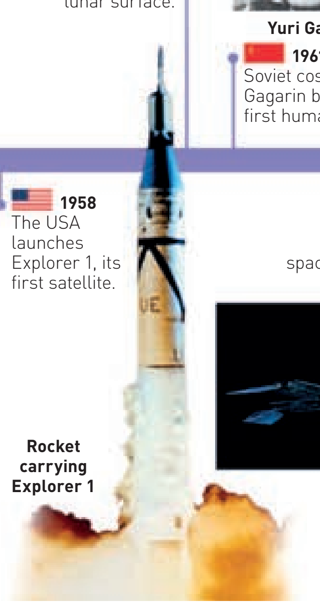
In 1957 the Soviet Union (USSR) launched a polished aluminium ball containing a temperature control system, batteries, and radio transmitter outside Earth's atmosphere. This was the beginning of the Space Age.

1950

1957 The Soviet Union marks the start of the Space Age when it launches Sputnik 1, the first man-made satellite.

Sputnik 1

1958 The USA launches Explorer 1, its first satellite.



Rocket carrying Explorer 1

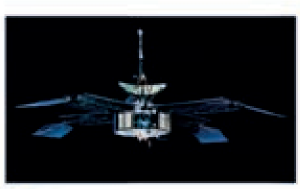
1959 The USSR launches Luna 2, which crashes on the Moon, becoming the first man-made object to reach the lunar surface.

Yuri Gagarin

1961 Soviet cosmonaut Yuri Gagarin becomes the first human in space.

1965 Soviet cosmonaut Alexei Leonov becomes the first person to perform a spacewalk.

1965 NASA's Mariner 4 becomes the first spacecraft to fly by Mars.

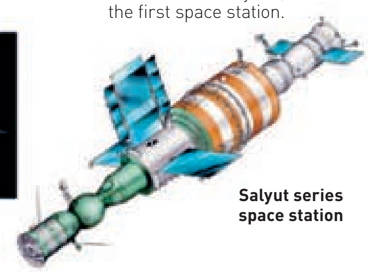


Mariner 4

1969 The USA's Neil Armstrong and Buzz Aldrin become the first humans to walk on the Moon.

Neil Armstrong

1971 The Soviet Union launches Salyut 1, the first space station.



Salyut series space station

1973 NASA's Pioneer 10 becomes the first spacecraft to travel beyond the Asteroid Belt and fly past Jupiter.

1973 NASA launches its first space station, Skylab.

Skylab

1977 NASA launches Voyager 1 and 2. Over the next few years they send images and scientific data from Jupiter and Saturn.

Voyager 1

1981 NASA launches Columbia, the first "space shuttle", or reusable spacecraft.